ETL MCQ Bank — Most Important & Weighted Edition

Focused questions with answers & explanations across: Introduction, Steps, Why ETL, Terminologies, Transformations, Informatica Mapping, ETL Testing, Process, Checklist, Roles, Challenges, ETL vs DB Testing.

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Legend:
• **MC** = single correct • **MS** = multiple correct. Attempt first, then check the explanation.

Prepared for: ETL/BI learners and interview candidates

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1. Introduction to ETL

Q1.

MC

What does ETL stand for?

- **A.** Extract, Transform, Load
- **B.** Evaluate, Test, Launch
- **C.** Export, Transfer, Link
- **D.** Encode, Transmit, Log

Answer: **A**

Explanation: ETL is the canonical pipeline: Extract data, apply business Transformations, then Load to target store.

Q2.

MS

Which are primary goals of ETL? (Select all that apply)

- **A.** Integrate data from multiple sources
- **B.** Improve data quality and consistency
- **C.** Load curated data into analytics storage
- **D.** Design front-end user interfaces

Answers: **A, B, C**

Explanation: ETL consolidates, cleanses and publishes data for analytics. UI design is out of scope.

Q3.

MC

Which layer typically receives the final ETL output?

- **A.** Staging area
- **B.** Operational source
- **C.** Data warehouse/mart/lakehouse
- **D.** Cache in application server

Answer: **C**

Explanation: Curated data is usually persisted in a warehouse, data mart or modern lakehouse table format.

Q4.

MC

Which statement about ETL vs ELT is correct?

- **A.** ETL transforms in the target
- **B.** ELT transforms in the target compute engine
- **C.** ETL never extracts data
- **D.** ELT cannot scale

Answer: **B**

Explanation: In ELT, data is loaded first then transformed using target compute (e.g., MPP SQL engines).

Q5.

MC

Which artifact captures business rules for ETL?

- **A.** Mapping/specification document
- **B.** CSS guidelines
- **C.** Sitemap
- **D.** Screen wireframes

Answer: **A**
Explanation: Mapping specifications drive transformation logic, column rules, and lineage.

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2. Steps of ETL

Q6.
MC

Correct order of ETL?

- **A.** Load -> Transform -> Extract
- **B.** Extract -> Transform -> Load
- **C.** Transform -> Load -> Extract
- **D.** Extract -> Load -> Transform

Answer: **B**
Explanation: Traditional ETL pipelines extract first, apply transformation logic, then load outputs.

Q7.
MS

Extraction typically includes: (Select all that apply)

- **A.** Incremental pulls using CDC/watermarks
- **B.** Full refreshes for small lookup tables
- **C.** Source profiling and data quality checks
- **D.** OLAP cube processing

Answers: **A, B, C**
Explanation: CDC and watermarks reduce volume; small lookups may be full; profiling supports downstream rules.

Q8.
MS

Transformation tasks commonly include: (Select all that apply)

- **A.** Standardization and cleansing
- **B.** Conformance across sources
- **C.** Derivations/aggregations
- **D.** UI layout changes

Answers: **A, B, C**
Explanation: Transformations harmonize data, derive measures and fix quality issues—not UI concerns.

Q9.
MS

Loading best practices include: (Select all that apply)

- **A.** Disable/enable indexes appropriately
- **B.** Batch/partition loads
- **C.** Upsert strategies for dimension/fact tables
- **D.** Skip constraints entirely

Answers: **A, B, C**
Explanation: Use bulk patterns, partitions, and upserts while preserving constraints.

Q10.
MC

Which artifact verifies row counts between source and target?

- **A.** Reconciliation report
- **B.** CSS audit
- **C.** Storyboard
- **D.** Ping test

Answer: **A**
Explanation: Basic sanity: row count reconciliation across pipeline legs.

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3. Why do we need ETL?

Q11.
MS

Which are benefits of ETL? (Select all that apply)

- **A.** Better decision-making via trusted data
- **B.** Data consolidation across silos
- **C.** Regulatory reporting readiness
- **D.** Game engine rendering

Answers: **A, B, C**
Explanation: ETL provides high-quality integrated data for BI, analytics, and compliance.

Q12.
MC

Which scenario most strongly justifies ETL?

- **A.** Multiple heterogeneous sources feeding an enterprise warehouse
- **B.** Single CSV read once
- **C.** Frontend CSS optimization
- **D.** Local spreadsheet formatting

Answer: **A**
Explanation: ETL shines when integrating many systems to an enterprise repository.

Q13.
MS

Which non-functional requirements are improved by mature ETL? (Select all that apply)

- **A.** Traceability/lineage
- **B.** Repeatability and scheduling
- **C.** Data quality SLAs
- **D.** 3D rendering speed

Answers: **A, B, C**
Explanation: ETL platforms emphasize control, lineage, and reliable scheduling.

Q14.
MC

Which metric best captures business trust in ETL outputs?

- **A.** Data reliability/defect rate
- **B.** Browser FPS
- **C.** CSS LCP
- **D.** Pixel density

Answer: **A**
Explanation: Downstream trust improves as data defects decline.

Q15.
MC

Which is a typical trigger to implement ETL?

- **A.** Merger/acquisition consolidating systems
- **B.** Color theme change
- **C.** Logo redesign
- **D.** Static brochure website

Answer: **A**
Explanation: M&A; drives system consolidation—classic ETL driver.

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4. Terminologies used in ETL

Q16.
MC

What is a staging area?

- **A.** Temporary landing zone before transformation
- **B.** OLAP cube
- **C.** Frontend cache
- **D.** NoSQL index

Answer: **A**
Explanation: Staging decouples sources and allows validation before heavy transforms.

Q17.
MS

Metadata includes: (Select all that apply)

- **A.** Data about data (lineage, definitions)

- **B.** Physical/technical metadata (schemas, types)
- **C.** Operational metadata (job runs, errors)
- **D.** CSS classes

Answers: **A, B, C**
Explanation: Business, technical, and operational metadata support governance.

Q18.
MC
Slowly Changing Dimension (SCD) Type 2 means:

- **A.** Overwrite attribute values
- **B.** Retain history with new rows and effective dates
- **C.** Delete old rows
- **D.** Store in cache only

Answer: **B**
Explanation: SCD2 tracks history via new records and date/version columns.

Q19.
MC
Conformed dimensions are:

- **A.** Shared, consistent dimensions across facts and marts
- **B.** Unique to each report
- **C.** Unrelated to star schema
- **D.** Only in OLTP

Answer: **A**
Explanation: Conformed dimensions standardize business keys across domains.

Q20.
MS
Surrogate keys are used because: (Select all that apply)

- **A.** Stable, system-generated identifiers
- **B.** Avoid dependency on volatile natural keys
- **C.** Support SCD strategies
- **D.** Improve UI theme

Answers: **A, B, C**
Explanation: Surrogates enable stable joins and historical tracking.

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5. Types of Transformations

Q21.
MS
Which are common cleansing transformations? (Select all that apply)

- **A.** Trim/case normalization
- **B.** Standardize codes (e.g., country)

- **C.** Null imputation
- **D.** UI layout refactor

Answers: **A, B, C**
Explanation: Cleansing fixes whitespace, casing, codes, and missing values.

Q22.
MS
Active vs passive transformations (ETL tools): which are active? (Select all that apply)

- **A.** Aggregator
- **B.** Filter
- **C.** Rank
- **D.** Expression

Answers: **A, C**
Explanation: Active transformations often change row counts or grouping; aggregator and rank are classic examples.

Q23.
MC
Which transformation implements SCD Type 1?

- **A.** Overwrite attributes in dimension
- **B.** Insert new row per change
- **C.** Flag deleted rows only
- **D.** Do nothing

Answer: **A**
Explanation: Type 1 updates in place—no history.

Q24.
MS
Which help detect duplicates? (Select all that apply)

- **A.** Window functions (ROW_NUMBER)
- **B.** Group by counts
- **C.** Fuzzy matching / soundex
- **D.** CSS cascade

Answers: **A, B, C**
Explanation: De-dup uses analytic functions, grouping, and fuzzy logic.

Q25.
MC
Which transformation derives aggregates for fact tables?

- **A.** Sum/avg group by keys
- **B.** Case normalization
- **C.** Regex tokenization
- **D.** Theme switch

Answer: **A**

Explanation: Aggregation derives measures at grain (e.g., daily, by customer).

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6. Mapping in Informatica

Q26.

MC

An Informatica mapping primarily defines:

- **A.** Flow of data from sources through transformations to targets
- **B.** Look-and-feel of dashboards
- **C.** VPN configuration
- **D.** Email templates

Answer: **A**

Explanation: Mapping is the core ETL design artifact in Informatica PowerCenter.

Q27.

MS

Reusable transformations provide: (Select all that apply)

- **A.** Single-definition used across mappings
- **B.** Centralized maintenance
- **C.** Consistent logic reuse
- **D.** UI palette themes

Answers: **A, B, C**

Explanation: Reusable components speed maintenance and ensure consistency.

Q28.

MS

Sessions and workflows are used to: (Select all that apply)

- **A.** Execute mappings
- **B.** Schedule/sequence tasks
- **C.** Control dependencies
- **D.** Compile CSS

Answers: **A, B, C**

Explanation: Sessions run mappings; workflows orchestrate and schedule.

Q29.

MS

Which is a best practice for mapping performance? (Select all that apply)

- **A.** Pushdown optimization when feasible
- **B.** Partitioning large flows
- **C.** Minimize datatypes casts and unnecessary ports
- **D.** Disable constraints permanently

Answers: **A, B, C**

Explanation: Exploit database compute, partition, and keep flows lean.

Q30.

MC

Parameterization helps because:

- **A.** Promotes environment portability/dev-test-prod
- **B.** Hardcodes all values
- **C.** Prevents reuse
- **D.** Blocks scheduling

Answer: **A**

Explanation: Parameters externalize environment-specific settings.

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7. ETL Testing

Q31.

MS

Primary objective of ETL testing is to: (Select all that apply)

- **A.** Validate mapping rules
- **B.** Ensure data completeness/accuracy
- **C.** Verify performance within SLAs
- **D.** Change brand colors

Answers: **A, B, C**

Explanation: ETL testing focuses on correctness, completeness and non-functional thresholds.

Q32.

MC

Which testing type verifies column-level derivations?

- **A.** Transformation rules testing
- **B.** Navigation UI testing
- **C.** CSS testing
- **D.** Usability testing

Answer: **A**

Explanation: Column formulas, data type conversions, and calculations are validated here.

Q33.

MS

Incremental load testing ensures: (Select all that apply)

- **A.** Only new/changed data is processed
- **B.** Watermarks/CDC are correct
- **C.** No duplication or data loss
- **D.** New color scheme applied

Answers: **A, B, C**

Explanation: Verifies correctness of delta ingestion and idempotency.

Q34.

MS

Data quality checks typically include: (Select all that apply)

- **A.** Null handling rules
- **B.** Domain/value set conformance
- **C.** Outlier thresholds
- **D.** CSS reset

Answers: **A, B, C**

Explanation: DQ validation protects integrity and usability.

Q35.

MC

Referential integrity testing ensures:

- **A.** No orphan foreign keys in facts
- **B.** Dimensional relationships intact
- **C.** Violated FKs are flagged/handled
- **D.** More stylesheets load

Answer: **A, B, C**

Explanation: Facts should reference valid dimension keys; exceptions must be treated.

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8. ETL Testing Process

Q36.

MC

Typical first step in ETL testing is to:

- **A.** Analyze mapping/spec documents
- **B.** Change logo
- **C.** Write CSS
- **D.** Ship to production

Answer: **A**

Explanation: Understanding rules precedes query design and execution.

Q37.

MC

Reconciliation tests verify:

- **A.** Row counts between points/legs
- **B.** Mobile UI gestures
- **C.** Browser cache
- **D.** SMTP latency

Answer: **A**

Explanation: Count reconciliation catches gross defects before deep diffs.

Q38.

MS

Defect lifecycle in ETL testing typically includes: (Select all that apply)

- **A.** Log defect with evidence
- **B.** Assign to ETL dev or data engineer
- **C.** Fix, re-run, and close after validation
- **D.** Ignore if minor

Answers: **A, B, C**
Explanation: Evidence-based triage and retest close the loop.

Q39.
MS

Test data management best practices include: (Select all that apply)

- **A.** Subset anonymized production-like data
- **B.** Deterministic data for repeatability
- **C.** Versioned datasets
- **D.** Hardcode secrets in SQL

Answers: **A, B, C**
Explanation: Manage secure, repeatable, representative datasets.

Q40.
MC

Exit criteria for ETL testing commonly includes:

- **A.** Coverage achieved, critical defects resolved, metrics within SLA
- **B.** Chosen color theme approved
- **C.** All UI testcases executed
- **D.** No lineage required

Answer: **A**
Explanation: Completion needs quality gates, not UI signoffs.

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9. ETL Testing Checklist

Q41.
MS

Which items are essential in an ETL test checklist? (Select all that apply)

- **A.** Schema and data type validation
- **B.** Null/defaults and domain checks
- **C.** Key constraints and dedup rules
- **D.** CSS z-index

Answers: **A, B, C**
Explanation: Checklist addresses schema, content, and relational integrity.

Q42.
MS

Performance items to monitor include: (Select all that apply)

- **A.** Throughput and job duration
- **B.** Resource utilization (CPU/mem/I/O)
- **C.** Skew/partition imbalance
- **D.** Button hover state

Answers: **A, B, C**
Explanation: Non-functional health ensures reliability at scale.

Q43.
MS
Security/compliance checks include: (Select all that apply)

- **A.** PII masking or tokenization
- **B.** Access controls to staging/warehouse
- **C.** Audit trails and lineage
- **D.** Font license

Answers: **A, B, C**
Explanation: Data protection and traceability are core.

Q44.
MS
Error handling checks verify: (Select all that apply)

- **A.** Rejected rows captured with reasons
- **B.** Retry logic/idempotency on failures
- **C.** Alerting/notifications
- **D.** Gradient backgrounds

Answers: **A, B, C**
Explanation: Robust pipelines surface and manage bad records and transient issues.

Q45.
MC
Deployment validation includes:

- **A.** Config/parameter correctness across envs
- **B.** Driver/JDBC connectivity tests
- **C.** Schedule/calendar verification
- **D.** CSS cascade

Answer: **A, B, C**
Explanation: Post-deploy smoke validates configuration and connectivity.

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10. ETL Testing Roles and Responsibilities

Q46.
MS
ETL tester typically performs: (Select all that apply)

- **A.** SQL validation of mappings
- **B.** Design test scenarios/cases
- **C.** Defect reporting/triage
- **D.** Corporate branding

Answers: **A, B, C**
Explanation: Core role: author SQL, validate, and report issues.

Q47.
MS
Data engineer/ETL developer typically: (Select all that apply)

- **A.** Implements mappings/workflows
- **B.** Optimizes performance
- **C.** Resolves defects
- **D.** Manages BI storytelling

Answers: **A, B, C**
Explanation: Developers implement and optimize pipelines; BI narration is separate.

Q48.
MS
Business analyst contributions include: (Select all that apply)

- **A.** Define rules/acceptance criteria
- **B.** Review mapping specs
- **C.** UAT signoff
- **D.** Write kernel drivers

Answers: **A, B, C**
Explanation: BAs define rules and accept outputs; UAT signoff may involve them.

Q49.
MS
Test manager focuses on: (Select all that apply)

- **A.** Planning/estimations
- **B.** Environment and data readiness
- **C.** Risk reporting
- **D.** CSS coding

Answers: **A, B, C**
Explanation: Management ensures readiness, planning and risk tracking.

Q50.
MC
Who owns lineage and governance at scale?

- **A.** Data governance function
- **B.** Random developer
- **C.** Only UI team

- **D.** No one

Answer: **A**
Explanation: Governance centralizes policies for lineage, quality and access.

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11. ETL Testing Challenges

Q51.
MS
Key challenges include: (Select all that apply)

- **A.** Huge volumes/variety
- **B.** Complex business transformations
- **C.** Late arriving or slowly changing data
- **D.** Button hover timing

Answers: **A, B, C**
Explanation: Scale and complexity create testing difficulty.

Q52.
MS
Mitigations for volume challenges include: (Select all that apply)

- **A.** Sampling with risk-based full checks
- **B.** Parallelization/partitioning
- **C.** Query optimization/pushdown
- **D.** Removing constraints permanently

Answers: **A, B, C**
Explanation: Balance thoroughness with targeted full checks; scale with parallelism and pushdown.

Q53.
MS
For late-arriving facts, you should: (Select all that apply)

- **A.** Design compensation loads
- **B.** Use placeholder dimension members
- **C.** Reconcile after backfill
- **D.** Ignore them

Answers: **A, B, C**
Explanation: Compensate, use unknown members, and reconcile when data lands.

Q54.
MS
Common data quality issues are: (Select all that apply)

- **A.** Nulls/out-of-range
- **B.** Duplicate business keys
- **C.** Inconsistent encodings
- **D.** CSS prefixing

Answers: **A, B, C**
Explanation: DQs include missing values, duplicates, and encoding mess.

Q55.
MS
Environment challenges include: (Select all that apply)

- **A.** Test data refresh/obsolescence
- **B.** Missing drivers/connectivity
- **C.** Schedule collisions
- **D.** Font anti-aliasing

Answers: **A, B, C**
Explanation: Ops realities often block consistent test runs.

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12. ETL vs Database Testing

Q56.
MS
ETL testing focuses on: (Select all that apply)

- **A.** End-to-end flow across legs
- **B.** Transformation correctness
- **C.** Load strategies (upsert, SCD)
- **D.** CSS gradients

Answers: **A, B, C**
Explanation: ETL spans multiple systems and transformations.

Q57.
MS
DB testing typically focuses more on: (Select all that apply)

- **A.** CRUD, constraints, triggers
- **B.** Stored procedures/functions
- **C.** Transaction/locking behavior
- **D.** Data lineage catalog

Answers: **A, B, C**
Explanation: DB tests validate database-internal logic and ACID semantics.

Q58.
MS
A core difference is that ETL testing requires: (Select all that apply)

- **A.** Cross-system reconciliation
- **B.** Source-to-target mapping validation
- **C.** Scheduling/orchestration checks
- **D.** Browser automation

Answers: **A, B, C**

Explanation: ETL involves stitching multiple systems with orchestration.

Q59.

MC

Which is best to validate end-to-end correctness?

- **A.** Source-to-target SQL QueryPairs
- **B.** Only unit tests in DB
- **C.** Manual spreadsheet compare
- **D.** Change logo

Answer: **A**

Explanation: Source vs target SQL comparisons validate transformation results.

Q60.

MS

When comparing ETL vs DB testing, performance validation for ETL emphasizes: (Select all that apply)

- **A.** Throughput across legs
- **B.** Job critical path duration
- **C.** Skew/partition balance
- **D.** Index rebuild every minute

Answers: **A, B, C**

Explanation: ETL performance is pipeline-scale, not only single query runtime.

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